PROGRAM 4

Implement SQL Functions: Conversion Functions, and Date Functions.

```
CREATE TABLE Student
(
  Student_ID INT,
  Name VARCHAR(50),
  Department VARCHAR(30),
  Marks INT,
  DOB DATE,
  Email VARCHAR(50)
);

INSERT INTO Student VALUES
(101, 'Rohit', 'CSE', 85, '2005-05-10', 'Rohit@gmail.com'),
(102, 'Rahul', 'ISE', 75, '2004-08-15', 'rahul@gmail.com'),
(103, 'Ramya', 'CSE', 90, '2005-01-20', 'ramya@gmail.com'),
(104, 'Arun', 'ECE', 65, '2004-12-05', 'arun@gmail.com'),
(105, 'Anu', 'ISE', 80, '2005-03-25', 'anu@gmail.com');

SELECT * FROM Student;

SELECT Name, CAST(Marks AS CHAR) AS Marks_As_Text
FROM Student;

SELECT Name, CAST(Marks AS DECIMAL(10,2)) AS Marks_As_Decimal
FROM Student;

SELECT Name, DATE_FORMAT(DOB, '%d-%m-%Y') AS Converted_DOB
FROM Student;

SELECT NOW() AS Current_Date_Time;

SELECT CURDATE() AS Today_Date;
```

```
SELECT Name, DAY(DOB) AS Birth_Day
FROM Student;

SELECT Name, MONTH(DOB) AS Birth_Month
FROM Student;

SELECT Name, YEAR(DOB) AS Birth_Year
FROM Student;

SELECT Name,
    DATE_ADD(DOB, INTERVAL 10 DAY) AS Date_After_10_Days
FROM Student;

SELECT Name,
    TIMESTAMPDIFF(YEAR, DOB, CURDATE()) AS Age
FROM Student;
```

Output:

Student_ID	Name	Department	Marks	DOB	Email
101	Rohit	CSE	85	2005-05-10	Rohit@gmail.com
102	Rahul	ISE	75	2004-08-15	rahul@gmail.com
103	Ramya	CSE	90	2005-01-20	ramya@gmail.com
104	Arun	ECE	65	2004-12-05	arun@gmail.com
105	Anu	ISE	80	2005-03-25	anu@gmail.com

Name	Marks_As_Text
Rohit	85
Rahul	75
Ramya	90
Arun	65
Anu	80

Name	Marks_As_Decimal
Rohit	85.00
Rahul	75.00
Ramya	90.00
Arun	65.00
Anu	80.00

Name	Converted_DOB
Rohit	10-05-2005
Rahul	15-08-2004
Ramya	20-01-2005
Arun	05-12-2004
Anu	25-03-2005

Current_Date_Time
2026-10-02 11:58:19

```
+------------+
| Today_Date |
+------------+
| 2026-10-02 |
+------------+
```

```
+-------+-----------+
| Name  | Birth_Day |
+-------+-----------+
| Rohit |        10 |
| Rahul |        15 |
| Ramya |        20 |
| Arun  |         5 |
| Anu   |        25 |
+-------+-----------+
```

```
+-------+-------------+
| Name  | Birth_Month |
+-------+-------------+
| Rohit |           5 |
| Rahul |           8 |
| Ramya |           1 |
| Arun  |          12 |
| Anu   |           3 |
+-------+-------------+
```

```
+-------+------------+
| Name  | Birth_Year |
+-------+------------+
| Rohit |       2005 |
| Rahul |       2004 |
| Ramya |       2005 |
| Arun  |       2004 |
| Anu   |       2005 |
+-------+------------+
```

```
+-------+--------------------+
| Name  | Date_After_10_Days |
+-------+--------------------+
| Rohit | 2005-05-20         |
| Rahul | 2004-08-25         |
| Ramya | 2005-01-30         |
| Arun  | 2004-12-15         |
| Anu   | 2005-04-04         |
+-------+--------------------+
```

```
+-------+------+
| Name  | Age  |
+-------+------+
| Rohit |   21 |
| Rahul |   22 |
| Ramya |   21 |
| Arun  |   21 |
| Anu   |   21 |
+-------+------+
```

Program 5 : **Implement SQL Operators: Demonstrate the use of different types of operators in SQL with examples, including Arithmetic Operators, Logical Operators, Comparison Operators.**

```
CREATE TABLE Student
(
  Student_ID INT,
  Name VARCHAR(50),
  Department VARCHAR(30),
  Marks INT,
  Fees INT
);

INSERT INTO Student VALUES
(101, 'Agney', 'CSE', 85, 50000);

INSERT INTO Student VALUES
(102, 'Rahul', 'ISE', 75, 45000);

INSERT INTO Student VALUES
(103, 'Ramya', 'CSE', 90, 50000);

INSERT INTO Student VALUES
(104, 'Arun', 'ECE', 65, 40000);

INSERT INTO Student VALUES
(105, 'Anu', 'ISE', 80, 45000);

SELECT * FROM Student;

SELECT Marks + 5 AS Increased_Marks
FROM Student;

SELECT Marks - 5 AS Decreased_Marks
FROM Student;
```

```
SELECT Marks * 2 AS Double_Marks
FROM Student;

SELECT Marks / 2 AS Half_Marks
FROM Student;

SELECT Marks % 10 AS Remainder
FROM Student;

SELECT * FROM Student
WHERE Marks = 85;

SELECT * FROM Student
WHERE Marks > 80;

SELECT * FROM Student
WHERE Marks < 80;

SELECT * FROM Student
WHERE Marks >= 80;

SELECT * FROM Student
WHERE Marks <= 80;

SELECT * FROM Student
WHERE Marks <> 80;

SELECT * FROM Student
WHERE Marks > 70 AND Department = 'CSE';

SELECT * FROM Student
WHERE Marks > 85 OR Department = 'ISE';
```

```
SELECT * FROM Student

WHERE NOT Department = 'CSE';
```

Output:

Output:

Student_ID	Name	Department	Marks	Fees
101	Agney	CSE	85	50000
102	Rahul	ISE	75	45000
103	Ramya	CSE	90	50000
104	Arun	ECE	65	40000
105	Anu	ISE	80	45000

Increased_Marks
90
80
95
70
85

Decreased_Marks
80
70
85
60
75

Double_Marks
170
150
180
130
160

Half_Marks
42.5000
37.5000

```
|    45.0000 |
|    32.5000 |
|    40.0000 |
+------------+
+-----------+
| Remainder |
+-----------+
|         5 |
|         5 |
|         0 |
|         5 |
|         0 |
+-----------+
+------------+-------+------------+-------+-------+
| Student_ID | Name  | Department | Marks | Fees  |
+------------+-------+------------+-------+-------+
|        101 | Agney | CSE        |    85 | 50000 |
+------------+-------+------------+-------+-------+
+------------+-------+------------+-------+-------+
| Student_ID | Name  | Department | Marks | Fees  |
+------------+-------+------------+-------+-------+
|        101 | Agney | CSE        |    85 | 50000 |
|        103 | Ramya | CSE        |    90 | 50000 |
+------------+-------+------------+-------+-------+
+------------+-------+------------+-------+-------+
| Student_ID | Name  | Department | Marks | Fees  |
+------------+-------+------------+-------+-------+
|        102 | Rahul | ISE        |    75 | 45000 |
|        104 | Arun  | ECE        |    65 | 40000 |
+------------+-------+------------+-------+-------+
+------------+-------+------------+-------+-------+
| Student_ID | Name  | Department | Marks | Fees  |
+------------+-------+------------+-------+-------+
|        101 | Agney | CSE        |    85 | 50000 |
|        103 | Ramya | CSE        |    90 | 50000 |
|        105 | Anu   | ISE        |    80 | 45000 |
+------------+-------+------------+-------+-------+
+------------+-------+------------+-------+-------+
| Student_ID | Name  | Department | Marks | Fees  |
+------------+-------+------------+-------+-------+
|        102 | Rahul | ISE        |    75 | 45000 |
```

```
|        104 | Arun   | ECE        |    65 | 40000 |
|        105 | Anu    | ISE        |    80 | 45000 |
+------------+--------+------------+-------+-------+
+------------+--------+------------+-------+-------+
| Student_ID | Name   | Department | Marks | Fees  |
+------------+--------+------------+-------+-------+
|        101 | Agney  | CSE        |    85 | 50000 |
|        102 | Rahul  | ISE        |    75 | 45000 |
|        103 | Ramya  | CSE        |    90 | 50000 |
|        104 | Arun   | ECE        |    65 | 40000 |
+------------+--------+------------+-------+-------+
+------------+--------+------------+-------+-------+
| Student_ID | Name   | Department | Marks | Fees  |
+------------+--------+------------+-------+-------+
|        101 | Agney  | CSE        |    85 | 50000 |
|        103 | Ramya  | CSE        |    90 | 50000 |
+------------+--------+------------+-------+-------+
+------------+--------+------------+-------+-------+
| Student_ID | Name   | Department | Marks | Fees  |
+------------+--------+------------+-------+-------+
|        102 | Rahul  | ISE        |    75 | 45000 |
|        103 | Ramya  | CSE        |    90 | 50000 |
|        105 | Anu    | ISE        |    80 | 45000 |
+------------+--------+------------+-------+-------+
+------------+--------+------------+-------+-------+
| Student_ID | Name   | Department | Marks | Fees  |
+------------+--------+------------+-------+-------+
|        102 | Rahul  | ISE        |    75 | 45000 |
|        104 | Arun   | ECE        |    65 | 40000 |
|        105 | Anu    | ISE        |    80 | 45000 |
+------------+--------+------------+-------+-------+
```

PROGRAM 6

Implement SQL Operators: Demonstrate the use of di”erent types of operators in SQL with examples Special Operators, and Set Operations(UNION)

```
CREATE TABLE Student
(
  Student_ID INT,
  Name VARCHAR(50),
```

```
  Department VARCHAR(30),
  Marks INT,
  Email VARCHAR(50)
);

INSERT INTO Student VALUES
(101, 'Rohit', 'CSE', 85, 'Rohit@gmail.com');

INSERT INTO Student VALUES
(102, 'Rahul', 'ISE', 75, 'rahul@gmail.com');

INSERT INTO Student VALUES
(103, 'Ramya', 'CSE', 90, NULL);

INSERT INTO Student VALUES
(104, 'Arun', 'ECE', 65, 'arun@gmail.com');

INSERT INTO Student VALUES
(105, 'Anu', 'ISE', 80, NULL);

SELECT * FROM Student;

SELECT * FROM Student
WHERE Marks BETWEEN 70 AND 85;

SELECT * FROM Student
WHERE Marks NOT BETWEEN 70 AND 85;

SELECT * FROM Student
WHERE Department IN ('CSE', 'ISE');

SELECT * FROM Student
WHERE Department NOT IN ('CSE', 'ISE');
```

```
SELECT * FROM Student
WHERE Name LIKE 'A%';

SELECT * FROM Student
WHERE Name LIKE '%a';

SELECT * FROM Student
WHERE Email IS NULL;

SELECT * FROM Student
WHERE Email IS NOT NULL;

SELECT Name, Department
FROM Student
WHERE Department = 'CSE'

UNION

SELECT Name, Department
FROM Student
WHERE Department = 'ISE';
```

Output:

Output:

Student_ID	Name	Department	Marks	Email
101	Rohit	CSE	85	Rohit@gmail.com
102	Rahul	ISE	75	rahul@gmail.com
103	Ramya	CSE	90	NULL
104	Arun	ECE	65	arun@gmail.com
105	Anu	ISE	80	NULL

Student_ID	Name	Department	Marks	Email
101	Rohit	CSE	85	Rohit@gmail.com
102	Rahul	ISE	75	rahul@gmail.com
105	Anu	ISE	80	NULL

Student_ID	Name	Department	Marks	Email
103	Ramya	CSE	90	NULL
104	Arun	ECE	65	arun@gmail.com

Student_ID	Name	Department	Marks	Email
101	Rohit	CSE	85	Rohit@gmail.com
102	Rahul	ISE	75	rahul@gmail.com
103	Ramya	CSE	90	NULL
105	Anu	ISE	80	NULL

Student_ID	Name	Department	Marks	Email
104	Arun	ECE	65	arun@gmail.com

Student_ID	Name	Department	Marks	Email
104	Arun	ECE	65	arun@gmail.com

```
|        105 | Anu   | ISE        |    80 | NULL           |
+------------+------+------------+-------+----------------+
+------------+-------+------------+-------+-------+
| Student_ID | Name  | Department | Marks | Email |
+------------+-------+------------+-------+-------+
|        103 | Ramya | CSE        |    90 | NULL  |
+------------+-------+------------+-------+-------+
+------------+-------+------------+-------+-------+
| Student_ID | Name  | Department | Marks | Email |
+------------+-------+------------+-------+-------+
|        103 | Ramya | CSE        |    90 | NULL  |
|        105 | Anu   | ISE        |    80 | NULL  |
+------------+-------+------------+-------+-------+
+------------+-------+------------+-------+-----------------+
| Student_ID | Name  | Department | Marks | Email           |
+------------+-------+------------+-------+-----------------+
|        101 | Rohit | CSE        |    85 | Rohit@gmail.com |
|        102 | Rahul | ISE        |    75 | rahul@gmail.com |
|        104 | Arun  | ECE        |    65 | arun@gmail.com  |
+------------+-------+------------+-------+-----------------+
+-------+------------+
| Name  | Department |
+-------+------------+
| Rohit | CSE        |
| Ramya | CSE        |
| Rahul | ISE        |
| Anu   | ISE        |
+-------+------------+
```